Assignment: Linear/Non-linear Regression

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Academic Integrity Declaration

Check the statement that applies and sign your name.

Check	Statement
V	I have **NOT** used LLM (ChatGPT, etc), any online materials, or other students' reports/codes to complete this assignment.
	I have **NOT completed this assignment entirely on my own**.

If you received any assistance **that violates the class policy**, you must explain it in detail. A penalty will be applied depending on the **extent and nature of the assistance**.

Failure to disclose the assistance, you will get Grade F.

Examples:

- *Screenshots of LLM prompt,*
- *Explanation of assistance : who/how/what extent etc*

By signing below, I confirm that I have honestly and truthfully answered the academic integrity declaration.

Date: 2025.12.01

Sign: 유채은

What you need to submit

- **Submit**: Report & Source files as one ZIP file
  - Assignment_Curvefit_22200476.zip
- **Report:** pseudocode, output result and source code as instructed
- **Source code files:**
  - Assignment_Curvefit_22200476.cpp
  - myNP_22200476.h, myNP_22200476.cpp, myMatrix_22200476.h, myMatrix_22200476.cpp

Part 0: Tutorial

Review tutorial:

Part 1: linear regression

Z=[a1, a0]

Z=polyfit(T,P,1) // 에러를 최소화 하는 직선의 계수 a0, a1을 구함

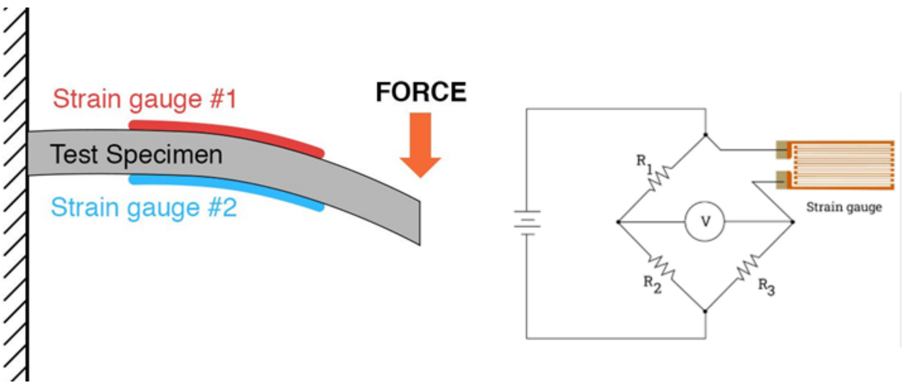
Part 2: higher order polynomial curve fitting
차수 n만큼 matrix 열이 늘어남
polyFit(Z_Q1, T, P, n);

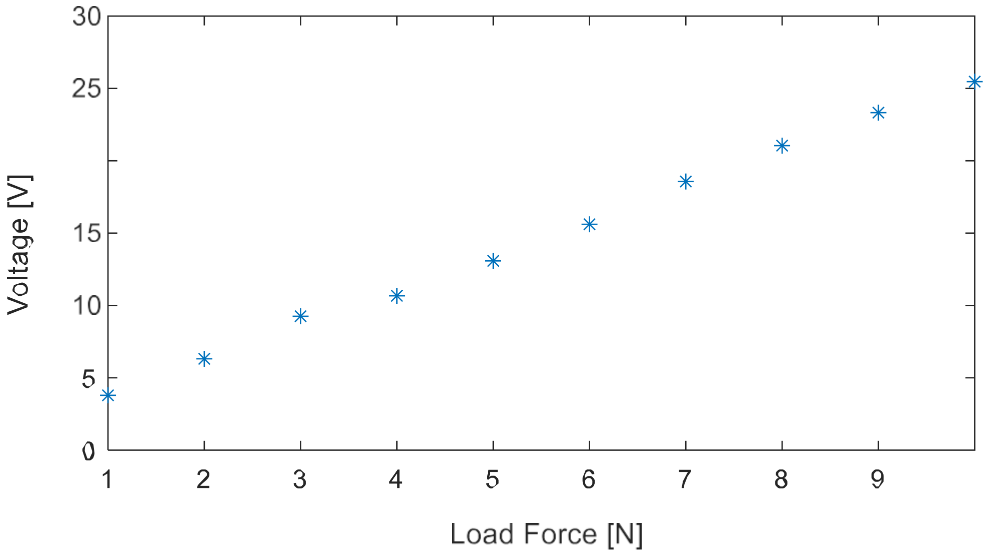
Part 1: Linear Regression

Download the template source file

Problem : Calibration of Strain-Gauge Experiment

An experiment is conducted to calibrate a strain-gauge sensor in a Wheatstone bridge. The measured voltage output V is assumed to have a linear relationship with the applied load force L:





$$V(L) = a_0 + a_1 L$$

a) Find the calibrated parameters for the strain gauge from experimental data.

b) Predict the voltage out when a new load is 11 N.

L= [1 2 3 4 5 6 7 8 9 10]

V =[3.7897 6.3118 9.2534 10.6665 13.0796 15.6108 18.5669 21.0461 23.3253 25.4691];

ANS:

Fitted line parameters: a0=____1.398____, a1=____2.421____

Estimate Voltage at L=10 N, V=____28.025____

Procedure

a) Write down a pseudocode for linear least squares regression

$$f(x) = \hat{y} = a_0 + a_1 x$$

```
for k = 0 to m
        Sx += x(k)
        Sxx += x(k)^2
        Sxy += x(k) * y(k)
        Sy += y(k)
End for

a0 = (Sxx * Sy - Sx * Sxy) / (m * Sxx - Sx * Sx);
```

```
a1 = (m * Sxy - Sx * Sy) / (m * Sxx - Sx * Sx);
```

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b) Create C/C++ function for Linear regression

```
void linearRegression (double z_opt[], double xdata[], double ydata[], int dataN) ;



// 1. Initialization
double Sx = 0;
double Sxx = 0;
double Sxy = 0;
double Sy = 0;
double a1 = 0;
double a0 = 0;

// Check m = length(X) and length(Y) ---> ?
// Is length(X)~= length(Y) ? Exit: Continue
// NOTE:
//             int mx = sizeof(vecX) / sizeof(vecX[0]); // <-- Does not work inside a
function

// Solve for Sx, Sxx, Sy, Sxy,
for (int k = 0; k < dataN; k++) {

      Sx += xdata[k];
      Sxx += pow(xdata[k], 2);
      Sxy += xdata[k] * ydata[k];
      Sy += ydata[k];
}


// Solve for a1, a2
a0 = (Sxx * Sy - Sx * Sxy) / (dataN * Sxx - Sx * Sx);
a1 = (dataN * Sxy - Sx * Sy) / (dataN * Sxx - Sx * Sx);

// Return z = [a0, a1]
z_opt[0] = a0;
z_opt[1] = a1;
```

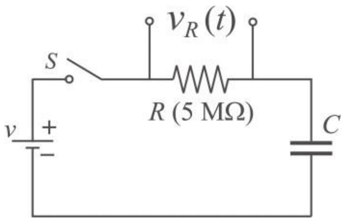
```
---------------------------------------------------------------
                 Part 1: Linear Regression (Line)
---------------------------------------------------------------
Z_Q1= [a0=1.398, a1=2.421]

Estimated V(L=11N)= 28.025
```

Part 2: Exponential Curve-fit $y = a_0 e^{a_1 x}$

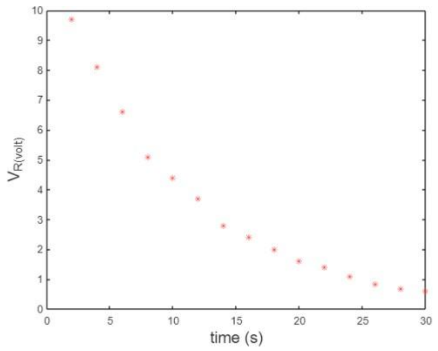
Problem :

Curve-fit the voltage across the resistor in 1st order RC circuit. How can you find the best parameters of capacitance C and initial voltage V_0?

Predict the voltage at t= 32 sec.



$$V_R(t) = V_0 e^{-t/RC}$$




```
Tdata = 2:2:30;
Vdata = [9.7, 8.1, 6.6, 5.1, 4.4, 3.7, 2.8, 2.4, 2.0, 1.6, 1.4, 1.1, 0.85,
0.69, 0.6];
```

ANS: RC circuit model parameter : C= 1.997_, V0= 11.913_

ANS: Voltage V(t) at t=32 sec: V= 0.483

Procedure

a) Write down a pseudocode for the exponential model least squares regression

---

```
for k = 0 to m
        Sx += x(k)
        Sxx += x(k)^2
        Sxy += x(k) * e^y(k)
        Sy += e^y(k)
End for


A0 = (Sxx * Sy - Sx * Sxy) / (m * Sxx - Sx * Sx);
a1 = (m * Sxy - Sx * Sy) / (m * Sxx - Sx * Sx);
```

b) Create C/C++ function for polynomial fit

```
void expRegression(double z_opt[], double xdata[], double ydata[], int dataN)
{
// 1. Initialization
double Sx = 0;
double Sxx = 0;
double Sxy = 0;
double Sy = 0;
double a1 = 0;
double A0 = 0; // A0 = ln(a0)

// Solve for Sx, Sxx, Sy, Sxy
for (int k = 0; k < dataN; k++) {

        Sx += xdata[k];
        Sxx += pow(xdata[k], 2);
        Sxy += xdata[k] * log(ydata[k]);
        Sy += log(ydata[k]);
}


// Solve for a1, a2
A0 = (Sxx * Sy - Sx * Sxy) / (dataN * Sxx - Sx * Sx);
a1 = (dataN * Sxy - Sx * Sy) / (dataN * Sxx - Sx * Sx);

// Return z = [a0, a1]
z_opt[0] = A0; // A0 = ln(a0)
z_opt[1] = a1;
}
```

```
-------------------------------------------------------------------
                   Part 2: Non Linear Regression (exp)
-------------------------------------------------------------------
Z_Q2= [a0=11.913, a1=-0.100]

Z_Q2= [C=1.997, V0=11.913]

Estimated V(t=32s)= 0.483

Press any key to continue . . .
```